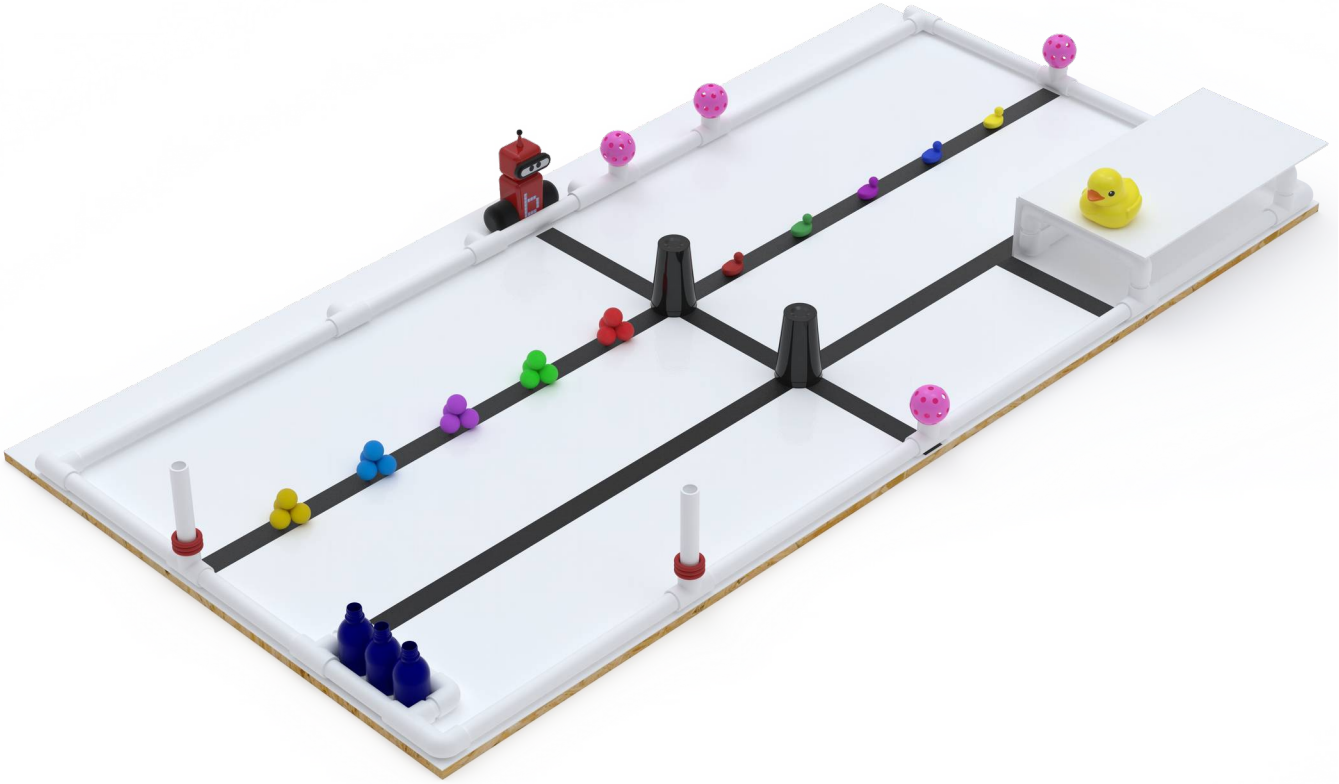
Fall Game 2026

2026 Botball Fall Game Review

Version 1.0



Contents

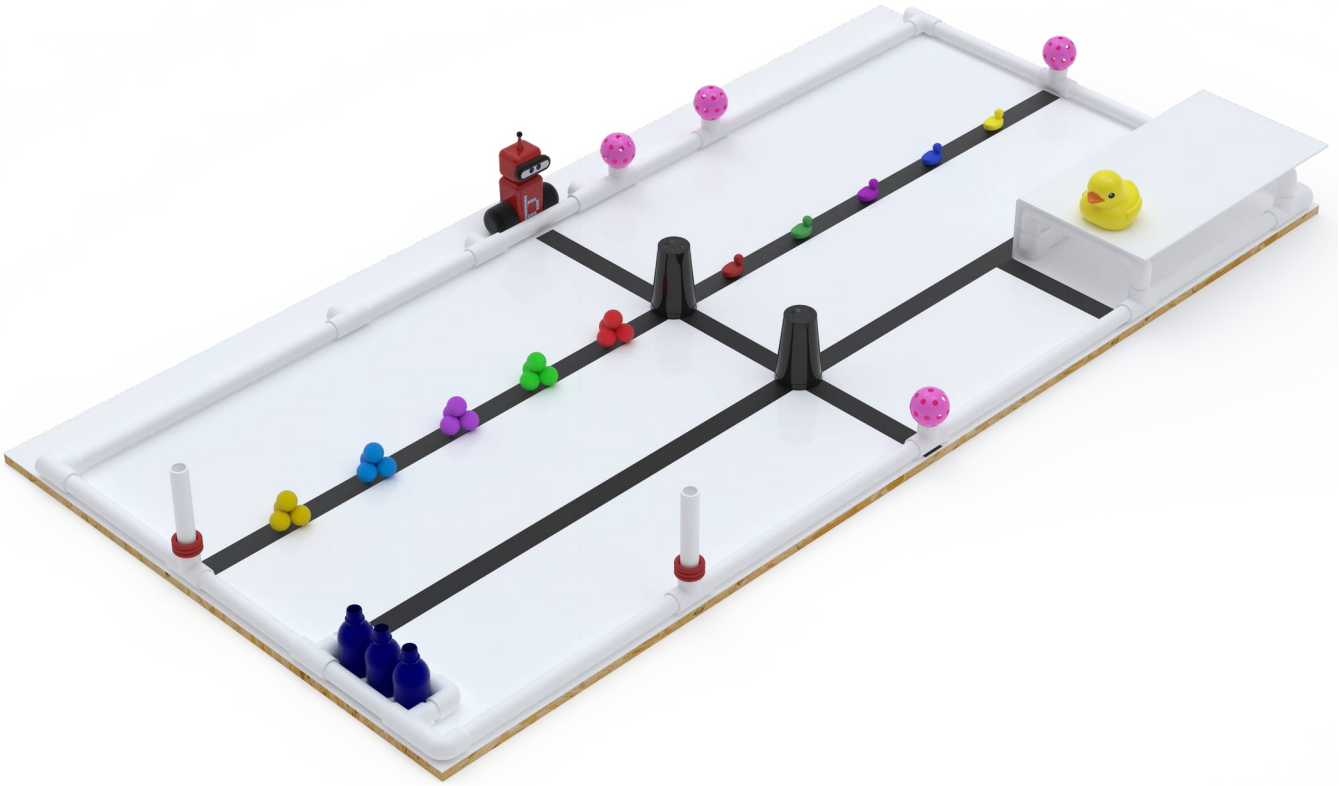
Virtual Fall Game....3
Scoring Pieces & Starting Positions....4
Scoring Rules....5
Definitions....7
Tie Breakers & Special Scoring Conditions....8
Gameplay....9
Tournament Structure....11
Construction Rules....11
Overall Winner Calculations....14

Virtual Fall Game

Teams earn points by participating in the seasonal game rules—such as sorting and moving designated objects and Botguy to different areas.

Virtual Tournaments requirements

- Participating teams must have a competition game board that meets the requirements specified in the table build document.
- Access to the internet
- Two cameras for competition day: one static overview camera showing the entire game board, and one movable camera for judge's robot inspection and point verification.
- Minor differences related to game board setup may be accommodated by the Head Judge at his or her discretion.
- An adult must be present for the entire tournament and will attest to the specification of the game board, robots, and game pieces being used.



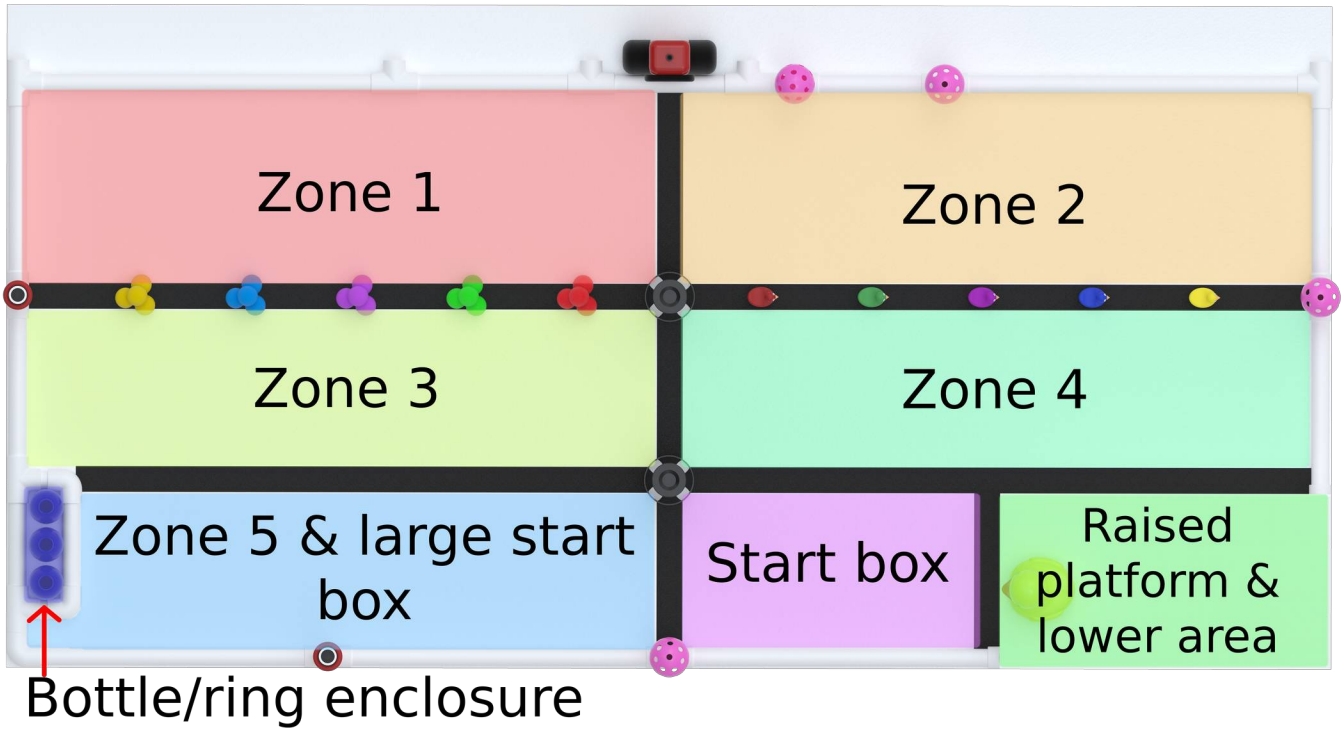
Game Board Areas

The board this year uses five black-tape-delineated zones, plus a Raised Platform, the area beneath it, and a separate Bottle/Ring enclosure.

- **Zone 1:** Top-left zone.
- **Zone 2:** Top-right zone.
- **Zone 3:** Middle-left zone (pom side).
- **Zone 4:** Middle-right zone (duck side).
- **Zone 5:** Lower-left zone — doubles as the larger Starting Box.

Excluded from the Overall Multiplier count (see Scoring Rules): the Starting Box, the Raised Platform, the area beneath the Raised Platform, and the Bottle/Ring enclosure (a distinct area bordering, but not part of, zone 5).

The starting box is 11.25" × 21.125" with a virtual height limit of 12".



Scoring Pieces & Starting Positions

Pieces

- **Botguy:** 1 piece. Starts centered left-to-right at the far top edge of the table.
- **Big Duck:** 1 piece, distinct from and larger than the small ducks. Starts on top of the Raised Platform.
- **Rubber Ducks (small):** 5 pieces, 5 colors.
- **Poms (visible):** 5 colors × 4 each, arranged in stacks of 3 with 1 on top, positioned along the center lane.
- **Poms (hidden):** 5 pieces, 1 of each color, hidden beneath the two cups, which start flipped upside down (2 poms under one cup, 3 under the other).
- **Mystery Red Pom:** The hidden red pom must be freed from its starting group to score its special bonus (see Scoring Rules). The red pom will be placed with two other poms in the cup closest to the starting box.
- **Cups:** 2 pieces (green, teal). Start upside-down, each hiding a group of the hidden poms.
- **Blue Bottles:** Located in the bottle/ring enclosure that borders zone 5.
- **Rings:** 6 total (3 per post, 2 posts: the far-left post and the post near the blue bottles).
- **Pickleballs:** 4 pieces, mounted on posts.

Piece Summary Table

Piece	Count	Starts	Scores
Botguy	1	Top-center, far top edge	On top of Raised Platform — 15 pts
Big Duck	1	On top of Raised Platform	At Botguy's starting position — 15 pts
Rubber Ducks (small)	5	Center lane	Base 1 pt; under-platform 3 pts; multiplier zones

Piece	Count	Starts	Scores
Poms (visible)	20 (5×4)	Center lane, stacked 3+1	Base 1 pt; see Pom scoring algorithm
Poms (hidden)	5	Under the 2 cups	Must be freed to join visible stacks
Mystery Red Pom	1 (of the 5 hidden)	Under a cup	Isolated alone on platform — 50 pts (multiplied)
Cups	2	Upside-down, hiding poms	See Cup scoring — up to 60 pts
Blue Bottles	—	Lower-left	Rings score onto these
Rings	6	2 posts (3 each)	3 pts (touching) / 15 pts (neck volume)
Pickleballs	4	Mounted on posts	Under Raised Platform — 7 pts each

Scoring Rules

Foundational Rules

- **Black Tape Rule:** Any game piece touching black tape does not score, except cups (cups are exempt from this rule only for their Nesting and Flat/Upright bonuses — a cup touching tape still fails to earn Base Score).
- **General Scoring Rule:** A game piece must touch the surface of the scoring area to score, except pieces following the Volume Rule.
- **Volume Rule (Ring/Bottle only):** To score a ring on a bottle at the 15-point level, the bottle neck must break the interior volume of the ring.
- **Highest Scoring Rule:** A game piece can only score in one scoring area and will be scored as if it is in the highest scoring area, as determined by base score without multipliers.
- **Robot Rule:** A robot is defined minimally as a KIPR Robot Controller with at least two motors connected to it that leaves the Starting Box. A robot with 2 controllers counts as a single robot.
- **Final Scoring Rule:** The score is determined by final object location, not by how it got there. Judges will wait until any scoring objects still in motion have come to rest before scoring a game.

Base Score

Every scoring piece touching the board surface, off black tape, scores 1 point. This applies in addition to any special scoring described below — Base Score and special scoring are additive, not exclusive.

Pom Scoring

For each group of poms found together in a single zone, apply the following:

- Same color, sorted, 1–4 poms: linear — 1, 2, 3, or 4 points.
- Same color, sorted, all 5 poms of that color together (a Complete Set) and no other colors present: flat 25-point bonus. There are exactly 5 poms of each color (4 visible + 1 hidden), so no duplicate-color ambiguity is possible.

- All different colors present, no duplicates, n poms: n^2 points (e.g., 2 poms/2 colors = 4, 3/3 = 9, 4/4 = 16, 5/5 = 25).
- Mixed group (some duplicate colors, some unique, neither fully sorted nor fully unique): falls back to Base Score only — 1 point per pom, no additional bonus.

Mystery Red Pom

One of the 5 hidden poms is red. If that red pom, and only that pom, ends up isolated on top of the Raised Platform (no other poms with it), it scores a flat 50-point bonus. This bonus is multiplied by the Overall Multiplier (see below). Any other pom present with it voids the bonus — this discourages simply dumping all poms onto the platform.

Duck Scoring

- Base Score: 1 point per duck, off tape (as with any piece).
- Under the Raised Platform: 3 points per duck.
- Ducks do not break sorts and have no color-sorting bonus of their own.

Overall Multiplier

Each of the 5 numbered zones that contains exactly 1 duck (no more, no fewer) adds +1 to a single, game-wide multiplier. This multiplier applies to the entire round's score — the full Round Subtotal, including the Mystery Red Pom bonus.

- 0 or 1 qualifying zone: multiplier is 1x (no effect).
- 2 qualifying zones: 2x. 3: 3x. And so on, up to a maximum of 5x.

The smaller Starting Box, the Raised Platform, the area beneath the Raised Platform, and the Bottle/Ring enclosure do not count toward this multiplier.

Cup Scoring

- Touching black tape, no special condition: 0 points.
- Off tape, on board, no special condition: 1 point (Base Score).
- Nested (one cup inside the other's volume), regardless of tape: 20 points.
- Flat/upright on the board (per cup, regardless of nesting or tape): 20 points each.
- Both cups nested and both flat/upright at once: 20 + 20 + 20 = 60 points.

Cups are the one piece exempt from the Black Tape Rule, but only for the Nesting and Flat/Upright bonuses above — a cup on tape still earns 0 toward Base Score.

Ring & Bottle Scoring

- Ring touching the bottle, not on the neck: 3 points.
- Ring on the bottle neck (bottle neck breaks the ring's interior volume): 15 points.

Pickleball Scoring

- Under the Raised Platform: 7 points each.

Botguy & Big Duck

- Botguy on top of the Raised Platform: 15 points.
- Big Duck at Botguy's starting position: 15 points.

Scoresheet

Item	Area	Count	Pts.	Subtotal
1. Base Score	Objects touching board, off black tape		1	
2. Pickleball	Under raised platform		7	
3. Rings	Touching bottle, not over neck		3	
3. Rings	On bottle neck (breaking volume)		15	
4. Ducks	Under raised platform		3	
5. Cups	One nested in other (1=Yes, 0=No)		20	
5. Cups	Flat and upright on the base		20	
6. Botguy	On top of raised platform		15	
7. Big Duck	At Botguy's starting position		15	
8. Hidden Red Pom	Alone on raised platform		50	
9. Poms: Same color, 1–4 poms	Numbered zones		1	
9. Poms: All 5 of 1 color	Numbered zones		25	
9. Poms: All different colors	Numbered zones		count × count	
9. Poms: Any other mix	Numbered zones		1	

Subtotal _____

× Zones with exactly 1 duck _____

= Total _____

Definitions

Board & Zones

Unless otherwise noted, all zones are delineated by the inner edge of the black tape and the vertical projection of any PVC borders.

- **Black Tape:** The boundary marking that, when touched by a scoring piece (except cups' Nesting/Flat bonuses), disqualifies that piece from scoring.
- **Starting Box:** The two designated boxes (one larger, one smaller) robots must start within; robots must individually fit within the smaller box, though one may start in the larger.
- **Multiplier Zone:** One of the 5 numbered zones that, when containing exactly one duck, adds +1 to the Overall Multiplier.
- **Raised Platform:** The elevated scoring area where Botguy and the Mystery Red Pom can score. Delineated by the edges of the corrugated plastic.
- **Area Under the Raised Platform:** A separate scoring zone beneath the platform, where pickleballs and ducks score.
- **Bottle/Ring Enclosure:** The post area holding rings and blue bottles; distinct from, and bordering, but not part of, zone 5.

Pieces

- **Botguy:** The robot supervisor piece, starting at the top-center, far edge of the table.
- **Big Duck:** The distinct, larger duck piece, starting atop the Raised Platform.
- **Pickleball:** Scoring piece (4 total), mounted on posts.
- **Pom:** Small scoring piece; 5 colors of 5 each (4 visible + 1 hidden).
- **Mystery Pom:** The specific hidden red pom, starting beneath a cup.
- **Cup:** Scoring piece (green, teal); can nest inside another cup and/or stand flat/upright.
- **Blue Bottle:** Scoring piece; rings score by breaking its neck's interior volume.
- **Ring:** Scoring piece (6 total, 3 per post); scores by touching or capturing a bottle neck.
- **Rubber Duck:** 5 small colored ducks; contribute to multiplier zones and under-platform scoring.

Scoring Mechanics

- **Base Score:** The flat 1-point award for any piece touching the board surface, off black tape.
- **Nesting:** One cup placed inside another cup's volume.
- **Flat/Upright:** A cup resting with its bottom flat against the board surface.
- **Complete Set:** All 5 poms of a single color together, alone, in one zone.
- **Breaking the Volume:** When one piece's relevant feature passes into another piece's open interior (used directionally for rings/bottles).

Tie Breakers & Special Scoring Conditions

Two disqualifying conditions apply, in this order of priority:

1. A robot must leave the Starting Box during the round; a robot that never breaks the Starting Box boundary (including the 12" ceiling) loses / scores 0 for the round.
2. A robot must stop driving its motors and stop servo motion by the end of the round; failure to do so results in automatic disqualification for that round (see End of Game).

Seeding Rounds

Seeding is not head-to-head. Each team earns an individual score; there is no winner, loser, or tie.

Double Elimination Rounds (Non-Finals)

If two teams' scores are tied:

1. Replay the match once.
2. If still tied, work through this ordered tiebreaker list:
    - Most rings on bottle necks
    - Most complete pom sets
    - Most ducks counted toward the Overall Multiplier
    - Most cups flipped (flat/upright)
    - Most pickleballs moved (scored under the platform)
    - Scoring the red Mystery Pom
    - Closest to Botguy (measured from the nearest corner of the robot's Wombat controller to Botguy)

All tiebreakers are measured entirely within each team's own single-sided board — none depend on an opponent's robot or pieces being present.

Double Elimination Finals

Tiebreakers are not used in the Finals. The match will be replayed until one team scores strictly more points than the other.

Gameplay

Fair Play and Spirit of Botball

Botball is about the development of student skills. All adults supervising a team should understand that responsible Botball mentorship does not include working on the robot entries or programming the robot entries for the students but does allow for appropriate mentor guidance of the team. Virtual Events require an adult to be present for the entire tournament and who will attest to the specifications of the game board and game pieces being used.

> ***Spirit of Botball: This is a 100% student-driven experience. Students know this, and adults know better!***

Inspections

A virtual judge may ask for a part count or to view robots in a zoomed-in view on the camera. The mentoring adult/coach will help the judge with the inspection and attest to the compliance of the robot.

Setup — Before Hands-Off

- Up to two students from a team may bring the team's robot(s) to the tournament table and perform setup.
- Teams will have 90 seconds to complete setup and calibration, placing robot(s) within their Starting Box(es) as desired.
- Starting light rules: lights are positioned on the outside edge of the game table; teams must adjust sensor mounts to match; lights must be aimed at the team's sensors or the floor; lights may not break the vertical projection of the board inside its PVC boundary; two connected starting lights per team, one per controller; teams cannot touch starting lights or any part of the table after Hands-Off.
- Teams must notify table judges before the end of "Hands-Off" if they believe the table is not set up properly or if there is something else they need to adjust. The team will receive confirmation from the judge, make the adjustment, and then — Hands-Off!
- If judges determine a team is taking too long to calibrate, they will issue a warning and set a 60-second timer. A team not ready after 60 seconds may be disqualified from the round. Maximum setup time (extendable at judges' discretion) is 90 seconds.

Timeout

Each team is granted a single timeout for the entire tournament, functioning the same as an in-person timeout, but without a physical card — teams notify the judge when they wish to use it. Timeouts may only be requested while the team is at the table before "Hands-Off." Once granted, the team receives a 3-minute timeout, which may be used to make adjustments to their robot, but not to practice runs at the table — however, teams may practice the starting light sequence during this time. Teams are advised to save their timeout for the Double Elimination rounds, as Seeding rounds are the best 2 out of 3.

If a robot is deemed to have an illegal part during Seeding rounds, a team may use their timeout to remove the offending part. If a robot is deemed to have an illegal part during Double Elimination rounds, the robot will be disqualified.

If any parts are added or modified during the timeout period, the Head Judge may ask to have the robot re-inspected once the timeout ends.

Before the Game Begins — After Hands-Off

Once "Hands-Off" has been declared, team members will position themselves so as not to block the judge's or the overview camera's view of the game table. No part of a team's robot(s) may leave the Starting Box(es) until the round has begun. Movement is okay so long as the Starting

Box(es) boundary isn't violated. If a moving violation happens, the judges will call a fault on the team. Team members may not move the starting lights at any time after Hands-Off. Team members may not signal to their robots after "Hands-Off" to start their robots. If a team member at the table is wearing a shirt with large solid colors similar to game pieces on the table, they will be asked to crouch or reposition during the round so as not to interfere with the robot's camera.

Teams are responsible for controlling outside light sources at their home table to prevent accidental triggering of the light sensors before the round starts. This may include shielding light sensors from windows, room lighting, or other environmental sources as needed to ensure a reliable start.

After the Game Begins — Lights On

Once the starting lights have turned on, the round counts unless a judge rules otherwise. At the start of the game, the starting lights turn on and robots are then allowed to leave the Starting Box.

The round lasts two minutes (120 seconds). The lighting sequence is:

- 0 seconds: lights turn on; robots can leave starting boxes
- 5 seconds: lights turn off
- 120 seconds: game over; robots must turn off motors and freeze/power down servos

Because this is a virtual tournament, teams may use a simpler home light-triggering setup and are not required to have an end-of-round light warning (e.g., a blink sequence). The 120-second time limit is enforced regardless of whether a team's home lighting reflects the end of the round — teams are responsible for tracking the round timer independently to ensure their robot stops on time.

End of Game

Robots must stop driving their motors and stop servo motion by the end of the round. Failure to do so results in automatic disqualification for that round. Incidental motion from a servo holding a position under load is OK.

If a team never breaks the boundary of their Starting Box during the round, that results in a score of 0 for the round (Seeding) — see Tie Breakers & Special Scoring Conditions for the equivalent Double Elimination handling.

Scoring is based on the location of pieces at the end of the round, not how the pieces got there. Scoring takes place once the round has ended and all items have come to rest.

If all motion has stopped before 120 seconds, the judge may ask the team if their robot is done. If the team agrees, the round may be ended early.

Final Scoring and Rulings

If a team does not agree with the score as calculated, they must immediately notify the judge before leaving the table and before any items have been moved. If they do not agree with the judge's ruling, they may ask to speak with the Head Judge. The Head Judge will spend no more than 5 minutes on the decision. Teams must initial the score sheet before leaving the table, signifying acceptance of the ruling. If they do not agree with the ruling, the Head Judge can sign for the team to progress the event forward.

There are no instant replays. No external videos will be used for replay scoring. If a team disagrees with a judge's decision, only the two team members at the table may politely discuss the issue with the table judge and/or Head Judge. Issues with the scoring after the teams and judges have agreed may not be considered.

The Head Judge reserves the right to make rulings on specific rules or wording in the game review that will be in effect for the remainder of the tournament.

> ***Spirit of Botball: Mentors, spectators, and team members should respect teams' and judges' final decisions. A Head Judge may overrule any previous rule or loophole that they believe to violate the Spirit of Botball.***

Acknowledgements

The KIPR Robot Controller is a powerful device, but the use of threading can cause unpredictable results, such as the robot not stopping when utilizing the shutdown in function. Teams are encouraged to limit their use of threading and to make sure they take precautionary steps to stop their robot within the time limit of the game.

Tournament Structure

Seeding Rounds

There will be three Seeding rounds, single-sided. A team's Seed Score is the average of their best two of three runs. Seed scores of less than 0 are counted as 0.

Double Elimination (DE) Rounds

A team is out of the Double Elimination tournament when it has lost two games. Teams compete virtually and simultaneously on their own single-sided boards; scores are compared to determine the winner of each match.

Not part of this game: Documentation scoring, Double Seeding, Alliance Matches, and Tiered Double Elimination rounds.

Virtual Tournaments

Virtual Tournaments require the team to have a competition game board that meets requirements, access to the internet, and two cameras: one static overview camera showing the whole game board, and one moveable camera for robot inspection and point verification on each run. Some exceptions to rules related to game board setup may be accommodated by the Head Judge.

Construction Rules

The official construction rules for the 2026 Botball Fall Game consist of the latest revision of this 2026 Botball Fall Game Review document and any updated game rules posted on the FAQ.

Kit Rules

- Sensors from the 2017, 2018, 2019, and 2020-2026 kits may be used so long as they don't exceed the type or number in the 2026 kit.
- Teams may use two cameras.
- KIPR Metal Parts — Aluminum chassis (old KIPR chassis) is not allowed.
- Robots may be constructed out of any or all of this year's kit parts except: boxes, bags, wrapping or packing material, chargers, download cables, wrenches, screwdriver, and color stickers. The included camera is the only USB device that may be plugged into a robot during the game.
- Small removable mounting dots/strips (Glue Dots, UGlu, Scotch Restickable Dots/Strips, blue tack) may be used for construction. Hot melt glue or other adhesives are not allowed.
- Zip ties used solely for wire management may be used as needed, without limit. Zip ties used for any other purpose (structural, mounting, etc.) are limited to five (5) total. Teams

are limited to the standard 4" white zip ties included in the kit for both purposes; damaged ties may be replaced with ones of equivalent size (black or white).
- Up to 100cm of thread, string, or fishing line (max diameter 1mm, non-metallic only) may be used, except for offensive measures such as entanglement.
- Paper (max 20#), foam board, or corrugated plastic, up to one standard letter/A4 sheet each, may be added; must be black or white/grayscale only.
- Up to 10 standard office rubber bands (max size #19) may be used; may not be glued or melted.
- Up to 10 metal paper clips (1"–1.5") may be used; cannot be cut, broken, or plugged into wiring.
- Coins, up to 250 grams, may be used as a counterweight only.
- The light sensors in the kit do not require a light guide unless there is significant direct sunlight in the room.
- Lego parts cannot be physically modified, except threading axle holes with screws.
- Metal parts may not be cut or broken to a smaller size; only straps and plates may be bent. Brackets cannot be bent.
- Teams are limited to using only 4' of the ABS plastic rods they receive.

3D Print Rules

- Allowed filaments: PLA or PETG.
- All printed parts must be grayscale-colored, regardless of material.
- Parts are measured by their total exterior bounding box volume, calculated in whole-inch units only — each dimension (height, length, width) is rounded up to the next whole inch before multiplying.
  - Small parts #6 (60mm × 60mm × 20mm)
  - Mid-size parts #2 (220mm × 220mm × 80mm).
  - Large parts #1 (220mm × 220mm × 250mm).
- Anti-circumvention clause: Splitting, segmenting, or otherwise designing parts with the intent of circumventing these volume-based limits — including assemblies of smaller prints that combine (by tethering, adjacency, or assembly) into a functionally or physically monolithic part — will be evaluated by the Head Judge as a single combined part for rule purposes. Judges retain full discretion to disqualify or require modification of any print determined to violate the intent of this rule.
- A single part cannot exceed the print volume of an Ender 3 V3 SE (220mm × 220mm × 250mm).
- An STL file must be submitted for each part in use prior to the tournament, per the schedule posted on the Team Homebase.

Robot Logistics

- Each robot, if named, can only have a name (G-rated) approved by an adult team leader before the tournament.
- Multiple processors (such as two KIPR robot controllers) may exist on a single robot.
- You may only use the Wombat controllers.
- It is not necessary to use all the parts in a kit.
- Interim rule: every robot must fit within the footprint of the smaller Starting Box, 11.25" × 21.125" × 12".
- All elements of a team's entry must be within the volume of a Starting Box at game start.
- After the game starts, robots are allowed to expand in size.
- Starting light sensors may be shielded as needed; neither the sensor nor shielding may extend outside the Starting Box.
- All independent structures not under computer control should be clearly marked with the team's number.
- Robot teams can have a maximum of 4 independent structures on the game table at a time.

- The ABS Plastic Rod must be permanently affixed to a robot by at least one end and may not be used as a projectile or independent structure.
- No electrical modifications may be made to any KIPR robot controller, sensors, or motors, except battery substitution with one approved by KIPR.
- No wire extensions may be used except those provided in the kit.
- Offensive entanglement strategies are not in line with the Spirit of Botball and may be subject to disqualification.

Safety

- No untethered robot-launched projectiles, other than game pieces, are allowed.
- No tethered projectiles containing metal pieces are allowed.
- No metal pieces or wires are to be used in effectors that move or rotate at high speed.
- No metal protrusions are to be used that are likely to cause electrical or safety risks for other robots (including arms and projectiles).
- Judges will determine how safe a robot is.
- Electrical tape (black or white) may be used to cover metal pieces deemed a safety risk. Tape may not be used for structural purposes.
- If the Head Judge decides a robot is not considered safe, it will not be allowed to run until modified or will be removed from the table.

External Communication

- No external communications (e.g., IR, Bluetooth, wireless, or semaphores) may be used during tournament play, with the exception of robot to robot (for a team's own multiple robots).
- Teams found programming a robot while at the table or in the on-deck area during rounds, excluding during a timeout, will receive a DQ for the round.
- USB cables and chargers may not be used during game table tournament play.
- A team's robot controller may have WiFi turned on or off at the tournament.

Overall Winner Calculations

A team's overall score for this game is the sum of their Seeding and Double Elimination scores. Documentation and Double Seeding do not apply to this game. Maximum possible Overall Score: 2.000.

Seeding Scoring Formula

$$Seed\ Score = \left(\frac{3}{4}\right) \times \left(\frac{(n - Seed\ Rank + 1)}{n}\right) + \left(\frac{1}{4}\right) \times \left(\frac{Team\ Average\ Seed\ Score}{Max\ Tournament\ Seed\ Score}\right)$$

n = Number of teams at tournament or in bracket

Double Elimination Bracket Scoring Formula

$$Double\ Elimination\ Score = \frac{n - DE\ Rank + 1}{n}$$

n = Number of teams at tournament or in bracket